

STEAM MODEL BLUEPRINT

The deepest external articulation of the STEAM Center model: what students experience, what makes it work, and what it takes to bring it to a new community.

Companion to the [Industry Partner Guide](#)

Read this Blueprint to understand how the model works. Read the Industry Partner Guide for what partners must do, decide, build, fund, and coordinate.

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ABOUT THE BLUEPRINT

This Blueprint is a detailed external description of the STEAM Center model, designed to make the model's architecture, student experience, and conditions for implementation visible and understandable to others. It documents what defines the model, how its core components work together, the practices and structures that bring those components to life, the system-level conditions required to sustain them, and the intended outcomes for students. It serves as both a comprehensive description of the STEAM Center model and a practical reference for communities seeking to understand, invest in, partner with, or replicate it.

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The Model at a Glance

Layer	What it holds
Core components	Work-Based Learning; Project-Based Learning in CTE; Progress Check-Ins; Postsecondary Planning. These are the experiences every student moves through.
Graduate aims	Industry Skills, Professional Skills, Identity, Agency, and Social Capital: the five outcomes the model develops in every graduate.
Design principles	Whole Child Focus, Relevance, Connection and Community, Customization, High Expectations with Rigorous Learning, and Agency, <i>drawn from Transcend's Leaps for Equitable, 21st Century Learning</i> .
Enabling conditions	Curriculum and assessment systems, school culture, scheduling and transportation, adult roles, partnerships, facilities, technology, budget, and continuous improvement: the infrastructure that makes the experiences possible.

The model provides common practices and expectations, but its components are designed to flex so schools can adapt implementation to their schedules, staffing models, and learner needs without compromising the integrity of the experience. It supports adaptation, iteration, and growth over time, allowing a community to start small, build capacity, and deepen its career-connected learning ecosystem.

PART 1: OVERVIEW

ILLUSTRATIVE DAY IN THE LIFE

Detailed Narrative

The STEAM Center model is a cross-school, industry-connected learning center. It integrates with, rather than replaces, a student's traditional high school experience: students keep their core academics at their home high school and spend part of every school day in a shared, industry-embedded center built around career pathways.

What the model is

The model describes how schools can design and sustain career-connected learning by aligning instructional experiences, organizational structures, and external partnerships around a common set of principles. It operates alongside traditional high school coursework rather than functioning as a stand-alone program. Students continue to take core academic classes such as English, mathematics, science, and social studies at their home high schools while participating in pathway-based learning that connects those subjects to real industries and applications.

The model serves students in grades 11 and 12, the years when learners prepare to transition to college, careers, or other postsecondary pathways. Students apply to the program and must be on track toward graduation to participate. This requirement ensures they have the academic foundation to engage in advanced project-based learning, industry-aligned coursework, and work-based learning. Once accepted, students split their school day: part at their home school for core academics, part at the STEAM Center for pathway instruction, industry-informed coursework, and work with professionals in their selected field. This shared schedule keeps students connected to their home school community while giving them access to a specialized career-connected learning environment.

A shared responsibility between two schools

Because students are enrolled at both their home high school and the STEAM Center, the model depends on a deliberate partnership between the two settings. Home schools remain responsible for core academic coursework, graduation requirements, special education and related services, attendance of record, and overall enrollment. The STEAM Center is responsible for pathway instruction and progress toward certifications. A designated point of contact at each home school partners with the STEAM Center's family and partnership coordinators to resolve scheduling conflicts, exchange student updates, and co-plan supports so that no student falls between the two settings.

Career pathways organize everything

Career pathways anchor curriculum and instruction to authentic fields such as healthcare, technology, engineering, or creative industries. The specific pathways vary by school and region, but the model always aligns projects, skill development, credentials, and work-based learning within each pathway.

Because pathways organize the entire student experience, how students enter and move through them matters. Before committing, students explore pathways through application and recruitment, introductory experiences, and early career-awareness activities such as industry tours and panels. Students formally select a pathway as they enter the program in 11th grade and commit to that single pathway for two years, because depth is what makes the model work: sequenced projects, credentials, and a coordinated work-based learning progression. Students still encounter adjacent fields through cross-pathway showcases, shared professional-skills instruction, and industry events.

The workplace is the classroom

Students from multiple schools come together at a center situated within an industry-connected site. By co-locating within or directly connected to an industry context, such as a hospital, airport, manufacturing facility, or cultural institution, students experience professional environments as a consistent part of their education. Within the center, they learn in specialized spaces equipped with industry-grade tools, collaborate in environments designed to mirror professional settings, and interact with industry professionals as a routine part of the school day. The center provides resources, environments, and experiences that individual schools could not offer alone, while students stay connected to their home schools for the rest of their education.

Industry partners are embedded

Industry partnerships run through the model as a core component of learning. Partners inform project design, contribute real-world expertise, and provide mentorship, feedback, and applied learning opportunities. These partnerships let students engage with authentic problems, understand industry expectations, and build professional networks.

This Blueprint describes industry partnership from inside the student experience: what partners contribute to learning. The operational, partner-side detail lives in the companion **Industry Partner Guide**. The two documents describe the same model from different angles and are designed to be read together.



HOW THIS DIFFERS FROM A TRADITIONAL CTE PROGRAM

Unlike traditional Career and Technical Education (CTE) programs, the model positions workplace immersion as the primary learning environment rather than a culminating experience. Five differences define it:

- The center is physically embedded within an active industry campus, so students engage with workplace culture and professional expectations daily.
- Industry partners co-design curriculum, projects, credentials, and skill standards, going well beyond hosting internships.
- Multi-school enrollment is an intentional equity strategy: the center operates as a regional resource that expands access across schools rather than a program housed in one school.
- Student support is distributed across eight student-led check-ins each year facilitated by four adult role groups, so each student builds relationships with multiple trusted adults instead of one assigned advisor.
- Outcomes are measured through a broader set of indicators, including credential attainment, paid work-based learning hours, employer satisfaction, identity development, agency, and social capital, alongside graduation and postsecondary enrollment.

Core Components

The specific activities learners engage in and practices they experience as part of the model. This is just a brief description of each component, as Part 3 will provide a deep dive into each component.

Pathway Selection

Students are initially exposed to [pathways](#) during the application process, where they learn about five broad industry pathway options rather than choosing from all twelve available specialties. Once admitted, students participate in an exploration period at the start of the program ([example plan](#)) that allows them to learn more about the pathway options before making a final commitment. The matching process is highly effective, with more than 95% of students receiving their first-choice industry pathway and 90–95% receiving their preferred specialty area within that pathway. Students primarily remain within one pathway once they make their final selection. Fewer than 1% of students [change pathways](#) after the initial selection process. Pathways are further reinforced through Career and Financial Management seminars, work-based learning experiences, industry partnerships, and post-secondary planning activities that help students understand how their pathway connects to future opportunities.

The Project-Based Learning in CTE and Work-Based Learning Connection

Project-Based Learning (PBL) is how students learn the work through teacher-facilitated, standards- and credential-aligned projects in STEAM classrooms and labs. Work-Based Learning (WBL) is where students do the work, applying those skills in authentic industry settings through job shadows, mentorships, internships, and paid placements. The two are intentionally sequenced: PBL builds skills, WBL tests them in real-world settings, and those experiences strengthen future learning.



Work-Based Learning is an intentional set of experiences that connects learning to real-world contexts, enabling students to apply academic knowledge and technical skills in authentic professional environments. Rather than functioning as a standalone program, work-based learning is integrated into the instructional model of the STEAM Center and aligned to career pathways, ensuring that experiences build progressively as students deepen their interests, skills, and sense of purpose. In the STEAM model, “classroom learning” in this component refers to the pathway instruction and project-based learning that happen at the STEAM Center, built on the academic foundation students develop in their home-school core courses. Work-Based Learning is integrated into the STEAM Center’s pathway instructional model—not the home school’s general academic program

—so that industry experiences extend and apply what students are learning in their pathway rather than functioning as a separate add-on.

Through activities such as industry projects, [mentorship](#), [Career Day](#), site visits, [internships](#), and paid or unpaid placements, students engage directly with professionals and workplace expectations. These experiences help students understand how learning translates into practice, develop professional skills, and explore potential career options while still in school. Work-based learning is designed to be developmentally appropriate, with opportunities increasing in depth, responsibility, and independence over time.

The model supports flexibility in how work-based learning is implemented, allowing schools to adapt experiences based on local industry access, student readiness, and scheduling constraints. At the same time, shared goals and design principles ensure that all work-based learning experiences are purposeful, equitable, and connected to student learning outcomes. When implemented effectively, work-based learning strengthens career readiness, reinforces academic learning, and builds students’ social capital by expanding their networks and real-world exposure.

➔ Go deeper into the [Work-Based Learning component](#).



Project-Based Learning (PBL) in Career and Technical Education (CTE) is an instructional approach that centers learning around authentic, industry-aligned projects that mirror real-world challenges within a specific career pathway. Students build technical knowledge and skills by actively applying what they are learning to meaningful problems, products, or performances that reflect professional standards and workplace expectations.

In CTE settings, projects are intentionally designed to integrate academic content, technical skill development, and professional competencies. Students engage in extended cycles of inquiry, design, revision, and reflection, often informed by industry

input or real-world constraints. Through this process, learners develop not only pathway-specific expertise, but also transferable skills such as collaboration, communication, problem-solving, and self-direction.

Project-based learning in CTE is flexible by design, allowing schools and educators to adapt project scope, pacing, and delivery to fit local contexts, student needs, and industry partnerships. While projects may vary across pathways and settings, they are anchored in shared learning goals, performance expectations, and quality criteria. When implemented effectively, PBL in CTE deepens student engagement, strengthens career readiness, and creates a coherent bridge between classroom learning and work-based experiences.

➔ Go deeper into the [PBL in CTE component](#).



Postsecondary Planning is a structured, student-centered process that supports learners in exploring, preparing for, and navigating pathways beyond high school. It helps students make informed decisions about college, careers, training programs, and other postsecondary options by aligning their interests, strengths, and goals with concrete next steps.

Through advising, reflection, and exposure to postsecondary opportunities, students develop clarity about available pathways and the requirements associated with each. This includes understanding application processes, financial aid, credentialing, and workforce entry, as well as how their high school experiences, such as coursework, projects, and

work-based learning, connect to future goals. Postsecondary planning is not a one-time activity, but an ongoing process that evolves as students gain new experiences and insights.

The model emphasizes flexibility and personalization, allowing schools to tailor postsecondary planning structures to meet student needs and local contexts. At the same time, shared tools, milestones, and guidance ensure that all students receive consistent

support and equitable access to information and opportunities. When embedded across the student experience, postsecondary planning strengthens agency, builds confidence, and positions students to transition successfully into their chosen postsecondary pathways.

➔ Go deeper into the [Postsecondary Planning component](#).



Progress Check-Ins are Brooklyn STEAM's version of structured, student-led coaching conversations. Through Progress Check-Ins, students regularly engage in intentional reflection and planning conversations with an adult advisor, with the goal of building clarity, ownership, and momentum toward their academic, personal, and postsecondary goals. In these conversations, students do the primary sense-making—analyzing their progress, naming challenges, and identifying next steps—while advisors provide guidance, probing questions, and strategic support rather than direction.

When executed with fidelity, Progress Check-Ins lead to increased student agency; stronger self-direction and goal-setting habits; the development of professional skills such as reflection, communication, and self-advocacy; and more coherent, future-

oriented learning experiences. Over time, students learn to see themselves as active planners of their own growth, rather than passive recipients of grades or advice.

To enable this level of student ownership, schools intentionally shift how they approach advising and support—moving from adult-driven check-ins and reactive interventions to student-led, forward-looking conversations that are embedded into the rhythm of the school. This shift requires clarity around purpose, consistent structures, and tools that help students prepare for and lead meaningful conversations about their progress and goals.

➔ Go deeper into the [Progress Check-In component](#).

Design Principles

The model's design reflects [Transcend's Leaps for Equitable, 21st Century Learning](#), a public framework describing the shifts learning environments make to serve every student well. Six Leaps run through the model:

Leap	How the model supports it
Whole Child Focus	The model integrates rigorous academics with identity development, professional skill-building, and future planning. Project-Based Learning, real-world Work-Based Learning, individualized Progress Check-Ins, and structured Postsecondary Planning support students intellectually, socially, and emotionally, not just academically.
Relevance	Learning connects academic content to authentic industry challenges, career pathways, and students' personal aspirations. PBL and WBL make learning applicable to real-world contexts, while check-ins and postsecondary planning connect daily effort to long-term purpose.
Connection and Community	Progress Check-Ins institutionalize meaningful adult relationships, PBL fosters collaborative peer communities, and Work-Based Learning expands students' networks beyond school, creating a layered web of support.
Customization	Students pursue industry pathways aligned to their interests, receive individualized goal-setting and support through check-ins, and develop postsecondary plans tailored to their strengths, aspirations, and readiness.
High Expectations with Rigorous Learning	The model treats every student as capable of excellence by embedding rigorous, performance-based tasks in PBL, aligning experiences to industry standards through WBL, and holding students accountable to graduation and credential benchmarks through structured monitoring.
Agency	Students act as decision-makers in their learning and future pathways. Through choice in projects, responsibility in professional settings, reflection in Progress Check-Ins, and ownership of Postsecondary Planning, students actively shape their trajectory rather than passively comply.

Graduate Aims

Every component of the model builds toward the same five outcomes. Together they describe a graduate who knows their industry, works like a professional, understands who they are, directs their own path, and can activate a real network.

Aim	Definition	Example indicators
Industry Skills	Graduates demonstrate mastery of tools, technologies, certifications, and applied practices aligned to high-demand industries. These skills are co-developed with industry partners, informed by labor market data, and validated through real-world performance tasks, credentials, and portfolio artifacts. Students leave career-aware and career-capable.	- Students with 54+ hours of work-based learning - Students earning at least one industry credential - Students earning the CTE Technical Endorsement
<u>Professional Skills</u>	Graduates communicate clearly, collaborate effectively, manage time and projects, give and receive feedback, solve problems, and demonstrate reliability and initiative. These skills are practiced through internships, project-based learning, progress check-ins, and authentic team-based experiences.	- Students attaining a “3” in each skill domain on the MHA Labs Framework and the Employability Profile Rating - Professionals also assess students after
Identity	Graduates possess clarity about who they are, including their strengths, passions, cultural background, values, and aspirations. They understand how their lived experiences shape their perspective and how they want to contribute to their communities and the broader world.	<i>Student-level measures for this aim are in development.</i>
Agency	Graduates set goals, reflect on progress, advocate for themselves, navigate systems, and adapt to changing circumstances. They understand how to seek help, leverage feedback, and make informed choices about college, career, and life direction.	Students who regularly discuss and review goals through progress check-ins
Social Capital	Graduates build meaningful connections with mentors, industry professionals, peers, and postsecondary advisors. They understand how to activate these networks, ask for support, and access information that opens doors.	- Work-based learning hours earned by students - Students participating in at least three work-based learning activities

Evidence of Impact

The results below come from the flagship site. They are Brooklyn's numbers, offered as proof of what the model can produce when implemented in full.

Students graduate with credentials and experience that lead directly to jobs and college success

- For the class of 2024, 98.5% of graduating seniors (139 of 141) earned at least one industry credential.
- 71.6% (101 of 141) earned a CTE Technical Endorsement, extra recognition on a New York State diploma signaling mastery of industry standards.
- 79.4% (112 of 141) earned 54 or more hours of work-based learning, meeting the highest NYSED standard for CTE.
- In 2024-25, 90% of graduates left STEAM having earned at least three college credits.

STEAM prepares students to succeed in both college and career

- Annually, 100% of students receive one-on-one college and career advisement, and 100% complete a personalized postsecondary plan aligned to their interests, credentials, and financial goals.
- Students can earn college credit through articulation agreements with SUNY Buffalo State, SUNY Canton, SUNY Oswego, and Alfred State. In 2025, all students were enrolled in a dual credit program that let them earn college credit by completing coursework at a "C" level or higher.

STEAM dramatically expands access to paid, high-quality work experience

- For the class of 2024, 83% of seniors (118 of 141) completed at least one paid work-based learning experience, counting both internships and industry-based projects. By contrast, a 2018 [American Student Assistance survey](#) found that although 79% of high school students wanted work experience, only 2% had ever completed an internship.
- From opening through July 2025, STEAM students have earned over \$1.4 million from paid work-based experiences.

PART 2: CORE COMPONENT DEEP DIVE

Work-Based Learning

Work-Based Learning (WBL) connects learning to real-world contexts, enabling students to apply academic knowledge and technical skills in authentic professional environments. In this model it is not a stand-alone program: it is integrated into the STEAM Center's pathway instruction so that industry experiences extend and apply what students are learning.

Purpose

WBL at the STEAM Center is a structured developmental pathway that connects classroom learning to authentic industry experience, aligned to [New York State Work-Based Learning guidance](#). Rather than treating internships or career exposure as isolated events, the model intentionally [sequences WBL experiences](#) so students progressively deepen their engagement with professional environments while supported by educators and industry mentors. In this component, “classroom learning” refers to the pathway instruction and project-based learning at the STEAM Center, built on the academic foundation developed in home-school courses.

The approach aligns with the broader NYSED WBL framework but organizes experiences intentionally across three stages of professional growth, so opportunities increase in depth, responsibility, and independence over time.

Stage	What happens	Example experiences
1. Career Awareness	Students build an understanding of industries, roles, and professional expectations, creating context for their pathway learning.	• Industry guest speakers and professional panels • Career fairs and industry expos • Site visits and industry tours • Credential exploration and career interest assessments
2. Career Exploration	Students engage more deeply with professionals and workplace contexts, applying knowledge to real problems.	• Industry-informed projects connected to coursework • Informational interviews with professionals • Job shadowing • Mentorship from industry partners • Community-based or service learning projects
3. Career Development	Students produce meaningful work within professional environments, applying technical skills, professional habits, and critical thinking in authentic contexts.	• Internships with industry partners • Apprenticeship-style learning experiences • Community-based work programs • Entrepreneurship or client-based projects • Clinical or industry-specific experiences depending on pathway

What makes this approach distinct

WBL here is a coordinated system rather than a set of isolated activities. It is integrated into each career pathway so that industry engagement directly informs coursework. Industry partners do more than host occasional visits or internships: they shape projects, critique student work, and introduce authentic problems that mirror their fields. Students experience career-connected learning not as an add-on to school but as a core part of how their learning is structured.

What adults do:

- The STEAM Center's work-based learning coordinator coordinates partnerships, manages logistics, and sustains industry relationships in ways individual schools struggle to do alone. The center functions as both a physical site and a coordinating entity that organizes employer engagement, schedules experiences, and distributes opportunities equitably across participating schools and pathways.
- Educators and industry mentors support students at every stage, matching students to experiences aligned to readiness and interest, with intentional progression from exposure to mentored project to internship or advisory placement.
- Staff log all activities in a centralized tracking system so participation meets NYSED requirements and schools maintain clear records. Students reflect on their learning, document their contributions, and demonstrate growth in technical and professional skills.



Connection to the graduate aims



Professional Skills

Professional skills develop most powerfully in authentic work settings. In WBL, students must communicate with supervisors and teammates, manage time and deadlines, collaborate across differences, adapt to feedback, and demonstrate reliability and initiative. These behaviors are reinforced through [real accountability](#). Because the stakes extend beyond grades, professional habits become lived experiences rather than abstract expectations.



Industry Skills

Students use industry-standard tools, workflows, and practices, and receive feedback from professionals who hold them to external standards, not just classroom rubrics. Through internships, client projects, and industry-reviewed capstones, students demonstrate mastery through performance, turning industry knowledge into practiced competence.



Agency

Navigating unfamiliar environments requires students to ask clarifying questions, seek feedback, set professional goals, reflect on strengths and growth areas, and make informed decisions about career interests. Structured reflection and coaching shift students from compliance to ownership.



Identity

Real industry spaces help students explore what environments energize them, what roles align with their values, how their background shapes their perspective, and where they want to contribute. Representation and mentorship matter: encountering professionals who share aspects of their identity builds confidence and clarity about who they are becoming.



Social Capital

Students form relationships with industry mentors, supervisors, clients, and professional networks. They learn to maintain those relationships, ask for recommendations, and navigate professional systems, expanding access to internships, jobs, references, and information that would otherwise remain out of reach.

Key structures and linked resources

Structure	What it does	Resources
Industry-guided feedback	Employers, professionals, and WBL partners give students feedback throughout their experience, bridging the gap between classroom learning and real-world expectations and ensuring students develop the professional and technical competencies the workplace expects.	• Training Plan template
Tiered experiences	Students are matched to experiences aligned to readiness and interest, with intentional progression from exposure, to mentored project, to internship or advisory placement.	• WBL Opportunities Grid • Sample WBL sequence • Job shadowing materials
Signature experiences	Mentorship, Career Day, site visits, internships, and paid or unpaid placements put students in direct contact with professionals and workplace expectations.	• Mentorship materials • Career Day materials • Internship materials

Project-Based Learning in Career Technical Education (CTE)

Project-Based Learning (PBL) is the primary instructional approach through which students develop technical expertise and professional skills within their chosen pathway. Unlike traditional PBL models that simulate real-world challenges, PBL at the STEAM Center is grounded in authentic industry work. Students are not completing projects for school; they are producing work that reflects what the workplace expects.

Purpose

Projects mirror the actual tasks, standards, tools, and constraints of the field, and industry partners often inform or co-develop them. Students build technical knowledge and skills by applying what they are learning to meaningful problems, products, or performances that reflect professional standards. Along the way they develop transferable skills: collaboration, communication, problem-solving, and self-direction.

What students experience

Within each pathway, learning is organized around a sequenced set of industry-aligned projects that build in complexity over time. These projects connect directly to the skills required for industry credentials, internships, and postsecondary pathways, so project work is not just a vehicle for engagement: it is how students progress toward tangible career outcomes.

Projects are embedded within each pathway's broader curriculum and assessment system, not designed as stand-alone activities. Every program includes at least one project or portfolio, and some include both. Structures vary by pathway: some culminate in a showcase or capstone ([example](#)), while others develop over an extended period with checkpoints across marking periods. Together the projects form a developmental progression, culminating in a portfolio that demonstrates readiness for internships, employment, and postsecondary opportunities.

What adults do

- Teachers, who often blend industry and pedagogical expertise, use targeted mini-lessons and demonstrations to introduce key skills, followed by extended time for students to apply those skills within their projects. Learning is immediately contextualized and reinforced through application.
- Because learning takes place in a workplace immersion environment, students have ongoing access to professionals who provide feedback, mentorship, and validation. Students regularly present to industry audiences in public showcases, where they must meet professional standards and respond to critique.

How projects are designed and validated

Projects are planned through a structured curriculum design cycle grounded in workforce relevance and external validation. The process begins with labor market data and industry reports, which faculty use to develop curriculum proposals. An advisory council and self-study team review and refine those proposals before they move through the state Perkins program approval process. This review ensures alignment to industry standards, academic rigor, certifications, work-based learning opportunities, and authentic work products. The cycle repeats regularly to keep curriculum current with evolving industry needs.

Students are assessed not only on what they know but on their ability to apply skills in authentic contexts, collaborate, and meet professional expectations. Rather than simulating real-world experiences, learning is embedded directly within them.

Connection to the graduate aims



Professional Skills

Projects require collaboration, communication, problem-solving, and time management in ways that reflect professional expectations. These skills are explicitly taught, practiced, and assessed across the project cycle, including deadlines, team contributions, and feedback from instructors and industry partners.



Industry Skills

Students apply technical knowledge through sequenced, industry-aligned projects that mirror real workplace tasks, using industry tools, technologies, and processes and building toward credentials and portfolio artifacts judged against authentic performance criteria.



Agency

Students make decisions about products, processes, and presentation; set goals; persist through challenges; and improve their work through cycles of feedback and revision. Ownership over learning increases intrinsic motivation.



Identity

Sustained engagement in pathway work helps students see themselves as contributors within a field. Creating and presenting high-quality work builds confidence and connects their interests and strengths to future pathways.



Social Capital

Project feedback, showcases, and collaboration put students in sustained contact with industry professionals, building relationships, exposing students to workplace norms, and expanding their networks.

Key structures and linked resources

Structure	What it does	Resources
Industry co-designed projects	Each project begins with a driving question rooted in a real industry problem. Projects are co-designed with industry partners, typically led by the pathway teacher, and the deliverable mirrors a professional work product such as a pitch deck, prototype, or recipe.	• Industry-Based Project Handbook
Industry-guided feedback across the cycle	Industry partners critique student work at multiple points (launch, midpoint, final), not just at culmination. Feedback elevates quality and builds students' social capital with adults from their field.	• N/A feedback is unique to each project/work sample
Capstone and portfolio	Each pathway includes a capstone or culminating task, and every student maintains a portfolio of project work that travels into postsecondary applications.	• Capstone example

Progress Check-ins

Progress Check-Ins ensure every student is known, supported, and actively planning toward their future. Through structured, recurring one-on-one conversations, students reflect and set goals with an adult advisor. Students do the primary sense-making; advisors coach.

Purpose

Unlike traditional advisory or grading conferences, Progress Check-Ins are student-led and future-oriented. Students come prepared to discuss their progress, identify challenges, and reflect on their experiences across academics, projects, work-based learning, and postsecondary preparation. Advisors help students interpret their experiences, ask deeper questions, and identify meaningful next steps.

This structure makes guidance proactive rather than reactive. Embedded in the regular rhythm of the school, check-ins build professional skills such as reflection, communication, goal-setting, and self-advocacy. Over time, students develop the habits and confidence to navigate complex decisions about their education and careers, and learn to see themselves as active planners of their own growth rather than passive recipients of grades or advice.

What students experience

- Monthly student-led coaching conversations. Students meet one-on-one with an advisor eight times a year, leading the discussion on their academic progress, personal development, project work, and future goals while the advisor asks probing questions and supports next steps.
- Pre-reflection preparation. Before each check-in, students complete a short pre-reflection form prompting them to reflect on progress, challenges, goals, and experiences, so they arrive ready to lead and advisors arrive with insight into student needs.
- Goal setting and action planning. Each check-in ends with clear action steps for both student and advisor, spanning academics, professional skills, certifications, work-based learning, and postsecondary planning.

What adults do:

Advisors act as coaches rather than directors. After each check-in they record key takeaways, action steps, and relevant progress information, allowing staff to monitor growth over time and keep support proactive and coordinated across the school community. Responsibility for check-ins is distributed across four adult role groups (pathway teachers, counselors, school leadership, and postsecondary planning staff), so every student builds relationships with multiple trusted adults.

Connection to the graduate aims



Professional Skills

The format mirrors a professional performance review. Students practice discussing progress, articulating challenges, and identifying actionable next steps, building communication, goal-setting, reflection, and responsiveness to feedback.



Industry Skills

Students regularly review technical progress: certifications, project work, and work-based learning. Advisors help track milestones, identify next steps for skill development, and keep pathway coursework aligned to career interests and required credentials.



Agency

Students make decisions about products, processes, and presentation; set goals; persist through challenges; and improve their work through cycles of feedback and revision. Ownership over learning increases intrinsic motivation.



Identity

Students prepare through pre-reflection, lead the discussion, and co-create goals and action steps, developing self-direction, self-advocacy, and initiative in identifying the opportunities and supports they need.



Social Capital

Check-ins build relationships with multiple trusted adults across the school community, expanding students' networks of support and connecting them to resources, opportunities, and guidance that carry into college, careers, and beyond.

Key structures and linked resources

Structure	What it does	Resources
Check-in calendar and advisor preparation	A published calendar sets the cadence for each round; advisors review student pre-flection data before every appointment and receive training on the coaching role.	• Monthly check-in calendar • Advisor training deck
Pre-flection forms and prompts	Industry partners critique student work at multiple points (launch, midpoint, final), not just at culmination. Feedback elevates quality and builds students' social capital with adults from their field.	• Reflection example
Documentation and progress tracking	Advisors log takeaways and action steps after each conversation so support stays coordinated across the school and patterns can be reviewed between rounds.	• Check-in logs and tracking tools

Postsecondary Planning

Postsecondary Planning is a structured, multi-year process that helps students explore, prepare for, and transition into pathways beyond high school: four-year and community colleges, trade schools, apprenticeships, military service, or direct entry into the workforce.

Purpose

Rather than focusing exclusively on college admissions, the model recognizes that students pursue a range of postsecondary routes. Planning begins with exploration and evolves into concrete preparation. Across grades 11 and 12, students clarify their interests, evaluate potential career pathways, and identify the education or training each requires. These experiences align closely with students' pathways, project-based learning, and work-based learning placements.

What students experience

Postsecondary planning is integrated across the model rather than treated as a separate counseling process. Students reflect during Progress Check-Ins, explore career options through work-based learning, and build evidence of their skills through projects and portfolios. During senior year, one day each week is repurposed from the pathway course for dedicated postsecondary planning time. Students receive guidance on:

- College and training program research
- Application timelines and requirements
- Financial aid and scholarship opportunities
- Credentialing and certification pathways
- Resume and portfolio development
- Professional networking and job search strategies

Each student graduates with an actionable [postsecondary plan](#), a resume, a professional portfolio, industry credentials, and at least one professional reference. Decision Day formalizes the plan publicly.

What adults do:

Postsecondary planning is a shared responsibility carried by counselors, school leaders, and program staff, not a stand-alone senior-year task. Counselors play the central role in helping students understand graduation requirements, application timelines, financial aid processes, and postsecondary options aligned to their interests and CTE pathways. Teachers and administrators use progress check-ins, student conferences, and day-to-day relationships to surface students' goals, barriers, and changing aspirations, while work-based learning lets students test those aspirations against real workplace expectations.

Families are part of the process. The center brings families into postsecondary planning through advising meetings, informational events, and communication about opportunities, deadlines, and requirements, so planning does not happen in isolation from home. For students navigating complex decisions about college enrollment, workforce entry, or technical pathways, this family-inclusive structure ensures they have both the information and the adult support to pursue ambitious next steps.

WHY THIS MATTERS

Students make better-informed decisions about life after high school because their plans are grounded in both aspiration and experience. Rather than only hearing about possible futures, students see pathways in action through workplace exposure, credential-bearing CTE programs, and conversations with adults who understand both the graduation path and the demands of college and careers.

Connection to the graduate aims



Professional Skills: Students practice resume writing, interview preparation, networking, and professional communication.



Industry Skills: Students align pathway coursework, industry credentials, and work-based learning experiences with postsecondary training or career opportunities.



Agency: Students research programs, complete applications, and make informed decisions about their future pathway.



Identity: Students reflect on their interests, strengths, and values to determine which postsecondary pathways align with their goals.



Social Capital: Students build relationships with counselors, industry partners, college representatives, and workforce organizations who support their transition after graduation.

PART 3: THE SYSTEM BEHIND THE EXPERIENCE

The student experiences described in this Blueprint depend on a set of enabling conditions that schools and the STEAM Center must intentionally build, staff, schedule, and resource. These conditions are essential design requirements for implementing the model effectively. Each community should determine how best to establish them within its local context. Where implementation depends on industry partners, the Industry Partner Guide outlines partner roles, expectations, and contributions in greater detail.

Curriculum, instruction, and assessment

Every pathway runs an industry-validated curriculum developed in house with support from industry partners.

- **Curriculum development:** Each pathway curriculum is built from labor-market data and industry input, reviewed by the [Industry Advisory Council](#) and a self-study team ([sample agenda](#)), and approved by the state through the Perkins process, which locks a program in for five years with amendments allowed when justified.
- **Shared across all pathways:** alignment to industry-relevant skills and stackable credentials; industry-grounded projects that build in complexity; explicit development of professional as well as technical skills; and industry professionals coming into classrooms while students go into industry settings.
- **Pathway-specific:** the technical content, tools, credentials, and culminating task. For example, the [Cybersecurity capstone](#) is assessed on two distinct skill sets: the technical work plus writing and slide design.
- **Instruction:** a guided, hands-on, project-centered approach built on a mini-lesson structure (a short teacher model or demonstration, then extended application), organized around an Essential Question, with intentional academic integration.
- **Assessment:** performance-based. Every program includes at least one project or portfolio, demonstrated publicly at twice-yearly [showcases](#) whose feedback feeds the next Advisory Council pathway review ([rubrics](#)). The model also tracks dual-credit attainment through articulation agreements and CTE Technical Endorsement attainment; each program offers at least three stackable credentials. See a [sample course syllabus](#).



School Culture

Culture is built deliberately around the idea that the school is a workplace, beginning the day students arrive and sustained through consistent relationships within the center.

- **Onboarding:** [Unit 0](#) is the foundational onboarding experience that moves students from new arrivals to emerging professionals by teaching the center's values, systems, and expectations.
- **Professional norms:** Students operate by workplace expectations, for example signing out equipment and leaving a workspace as they found it, reinforced consistently across pathways.
- **Relationships and trust:** The most formal structure is the monthly Progress Check-In, and a counselor meets with every student. The explicit aim is that each student builds at least one, and ideally two, trusted adult relationships.
- **Belonging across a multi-school center.** Because students come from 13 home schools in Brooklyn's implementation, community-building is intentional: an internal showcase in year one, social-emotional learning trips, weekend college visits, and twice-yearly mixers (a January carnival and a June games day) where students step out of their pathways to connect. Alumni events mirror the same cadence.
- **Discipline:** Handled through standard school processes: investigation and fact-finding, consultation with borough safety where appropriate, and mediation.

Scheduling and transportation

The model runs on a shared-day schedule split between students' home high schools and the STEAM Center. Two conditions are non-negotiable: the daily pathway block and midday transportation.

- **Shared day:** Students take core academics at their home school and have a [2.5-hour pathway block](#) at the center. In Brooklyn, seniors attend in the morning and juniors in the afternoon.
 - The 2.5-hour block is intentional. It provides sufficient seat time for entry-level career preparation and meets academic-policy requirements for shared instruction and CTE.
- **Layering the components:** Pathway project-based learning anchors the daily block. Work-based learning is scheduled into designated portions of the week, with counselor pull-outs as needed. Progress Check-Ins run during structured break periods on a flexible push-in/pull-out model: pathway teachers pull students individually during breaks, while administrators either join students where they are or pull them aside for a private conversation. In senior year, one day each week is repurposed from the pathway course for Postsecondary Planning.

- **Scheduled across the year in advance:** two annual showcases, January and June Advisory Council meetings timed to state testing weeks, work-based learning events, and pathway project cycles.
- **Transportation.** A coordinated midday shuttle, contracted in Brooklyn through the Office of Pupil Transportation, moves students between the 13 partner schools and the center. It is not an instructional feature, but it is essential for equitable participation in a distributed model.

Adult roles, hiring, and development

The model depends on a core teaching staff plus a set of coordination roles shared across the center, the anchor partner, and home schools.

- **Pathway/CTE teachers:** At least two qualified teachers per pathway, employed by the district, who blend industry and teaching experience. That blend makes both rigorous project-based learning and monthly Progress Check-Ins possible. In Brooklyn they are recruited through the [Transitional A certification](#) route and the [Success Via Apprenticeship program](#), supplemented by adjuncts hired directly from industry.
- **Work-Based Learning Coordinator:** Holds the NYSED WBL Extension Certificate; structures the program, recruits students and partners, maintains industry relationships, documents artifacts, and manages stipend payroll.
- **Counselor and Postsecondary Planning staff:** A counselor meets with every student; PSP staff own the senior-year planning process.
- **Family and partnership coordination:** A family or partnership coordinator and a Community Associate manage caregiver and community-based-organization relationships.
- **Anchor-partner roles:** A Director and an Internship Coordinator on the partner side plan workplace learning on the campus and place students; some roles are effectively co-shared between the anchor and the school ([role descriptions](#)).
- **Development:** Weekly professional development led by leaders and instructional coaches, plus ongoing industry currency maintained through Advisory Council interaction and short-term industry co-teachers.

Community and family partnerships

Three distinct partnership types sustain the model, and it helps to keep them separate: industry partnerships, family partnerships, and community-based student support.

- **Industry Advisory Council.** Industry professionals, together with higher-education and community partners, give ongoing input on curriculum, student work, and pathway development, and commit to attending meetings, participating in work-based learning, hosting interns, and completing an annual survey. State CTE approval sets a minimum composition: at least two industry representatives from separate organizations, at least one postsecondary partner, and student participants. See a [sample overview](#), [sample structure](#), and [sample meeting agenda](#).
- **Industry partners as curriculum, WBL, and feedback partners.** They co-design projects, supply authentic problems, and critique student work at showcases and throughout the project cycle.
- **Family partnerships.** Caregivers engage around postsecondary planning through advising touchpoints and informational events, and each partner school designates a caregiver coordinator who meets on the cadence set in the Memorandum of Partnership.
- **Community-based student supports.** Community-based organizations extend whole-child support beyond academics: for example, vision care, food access, and clothing. This support stream is distinct from the industry and family partnerships above.

Space and facilities

The defining facilities principle: the workplace is the classroom. The center is embedded inside an active industry campus rather than placed next to one.

- **Non-negotiable:** an industry-connected location that puts students in daily proximity to professionals, so a partner can step in to critique work and a student can head to an internship after class.
- **Specialized pathway spaces, designed with industry input.** Partners helped define specifications, review architectural plans, and select equipment so each environment mirrors its real-world counterpart. Brooklyn's spaces include a full sound stage with control room, editing booth, and screening room; a fabrication lab; a construction workshop; and a professional-grade kitchen.
- **Industry-grade tools throughout.** Students train on the same equipment professionals use, in an environment professionals regularly ask to borrow for their own work. See the [aesthetic of “the school is the workplace”](#).

Technology

Technology supports learning, operations, tracking, and communication throughout the center.

- **Core platforms:** a learning management system (Schoology in Brooklyn) and the Google Workspace suite, with 1:1 devices and reliable wireless access throughout the center.

- **Tracking systems:** digital portfolios and Employability Skills Profiles document student work and credentials; pre-flection surveys feed Progress Check-Ins; and work-based learning activity and stipends are tracked in the state STARS system and the Career Pathways Portal (CPP).

Budget and operations

Operations are shaped by co-location with the anchor partner and by funding streams that must cover staffing, stipends, equipment, dual credit, and transportation.

- **Funding streams in Brooklyn:** a tax-levy School Allocation Memorandum rather than per-pupil funding, supplemented by Perkins grants for work-based learning and equipment, philanthropy through the anchor partner, and other initiatives.
- **Student pay and stipends:** processed through the Career Pathways Portal, with the documentation and payroll steps paid work-based learning requires. For example, a Social Security number is required at or above the federal stipend threshold.
- **Procurement:** a defined process for purchasing CTE-specific equipment and consumables.
- **Costs to plan for:** dual-credit tuition, credentialing and exam costs, equipment and facility costs, and staffing and coordination, including the cost of an additional instructional period.
- Co-location implications: facilities, capital improvements, and logistics are coordinated jointly with the anchor partner.

Continuous learning and improvement

Improvement runs on a mix of recurring data routines, some systematic and some still developing..

- **Built-in cycle:** because Progress Check-Ins recur monthly, the pre-flection data and meeting records they generate are synthesized between rounds and reflected back to the team to adjust the next round.
- **Program-level routines:** an all-day systems stepback to evaluate programmatic elements, milestone check-ins with the district, and an annual planning process.
- **Data used:** credential, work-based learning, postsecondary, and student-progress data, plus Advisory Council feedback and student and partner input where collected. The model gathers data high schools do not typically track, such as workplace performance and social-capital growth, which requires purpose-built tools. A [pathway expansion process](#) guides the periodic stepback on what is missing or could grow.